



Original Article

Identification of Toscana virus in natural population of sand flies (Diptera: Psychodidae) from Moroccan leishmaniasis foci



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ABSTRACT

Background: Phlebotomine sand flies are known as vectors of various pathogens such as *Leishmania* sp parasite and Toscana virus (TOSV). Leishmaniasis is endemic in Morocco, and TOSV is increasingly reported. Our objective is to analyze the specific composition of the natural population of sand flies in endemic and non endemic area of leishmaniasis in Morocco, thus evaluated their infection by Toscana virus.

Methods: Sand flies were collected by CDC miniature light traps from seven different localities with an altitude range from 399 m to 1496 m. Synanthropic index was calculated for each sand fly species. The collected female sand flies were grouped in 73 pools, with a maximum of 50 specimens per pool, and submitted to real time PCR for TOSV detection.

Results: 8 sand fly species were identified morphologically: 5 of the *Phlebotomus* genus and 3 of the *Sergentomyia* genus. *Phlebotomus sergenti* was the most abundant species comprising of 43,12% of identified sand flies, followed by *P. papatasi* (18,89%) and *P. longicuspis* (13,43%). Estimated synanthropic indices for these species were between + 1.1 and + 12.6 suggesting a high preference to anthropogenic environments. A total of 3558 sand fly females were grouped in 73 pools (up to 50 sand flies per pool) for TOSV detection. TOSV was detected in one pool (out of 6 tested) from Lalla Laaziza locality (Chichaoua Province) where *P. sergenti* was the most abundant sand fly species.

Conclusion: We reported the TOSV for the first time in a central Morocco, where cutaneous leishmaniasis by *L. tropica* is endemic. This result has epidemiological importance for both researchers and health authorities to monitor circulation of TOSV and implement a surveillance plan of sand fly-borne phleboviruses in Morocco.

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Introduction

Phlebotomine sand flies (Diptera: Psychodidae, Phlebotominae) are small and fragile insects, they take blood meals from a range of animal hosts [1], which presents a zoonotic concern [2]. Sand flies are vectors of various agents of diseases, notably human and canine Leishmaniasis, bartonellosis, and several arboviruses [3].

Mediterranean countries saw an increase in cases of diseases linked to phleboviruses and the discovery of new viruses within this genus [4], many of which were transmitted by sand flies. *Phlebovirus* genus belongs to the family of *Phenuiviridae* (Order *Bunyavirales*) [5]; it is one of the most medically important groups of viruses for human and animal health [6]. Most phleboviruses are transmitted by sand flies, but there are others that are transmitted by mosquitoes [7].

Toscana virus (TOSV) is a phlebovirus causing disease in humans, mainly summer flu-like diseases, but infection can be associated with aseptic meningitis, meningoencephalitis (MMA) [8], and unusual cases of MMA, as reported in Italy and Spain [9,10]. In Europe TOSV was isolated from *P. perniciosus* [11] and *P. perfliewi* [12], while

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TOSV RNA was detected in *Sergentomyia minuta* in southeastern France [13].

Phleboviruses are widely circulating in Morocco. Tesh et al. [14] observed antibodies against the Naples virus and Sicilian virus in human populations living in Central Morocco. Like other vector-borne and zoonotic diseases, TOSV infections are the result of several interactions between different factors in space and time, as human behavior and environmental conditions [15]; the disease can be considered as a shift in the pathogen–host–environment interplay characteristics [16]. The detection of TOSV in Morocco, and subsequent circulation, poses a potential risk of associated neurological diseases. In 2012, TOSV RNA was detected in four pools of *P. perniciosus* for the first time in Morocco in the locality Louata in Sefrou province [17], two years later this virus was detected in Al Hanchane (Essaouira Province) district from *P. sergenti*, the main vector of *Leishmania (L.) tropica* in Morocco [18,19]. Toscana virus was then isolated from *P. sergenti* and *P. longicuspis* in Sefrou and Azilal provinces [20].

In Tunisia, the circulation of TOSV and *Leishmania sp* have been linked to environmental factors such as the intensive extension of irrigated surfaces in arid bio-geographical areas located in central Tunisia [21]. This caused significant environmental changes which caused the establishment of populations of sand flies of the subgenus *Larroussius* vectors of *L. infantum* and TOSV. These favorable conditions sustain the transmission cycles of *Leishmania* species and phleboviruses [21]. Given the presence of these data from other North African countries and regions of Morocco, there is a need to investigate the circulation of TOSV in Morocco.

Our objective is to analyze the specific composition of the natural population of sand flies in endemic and non endemic area of leishmaniasis in central Morocco, thus evaluated their infection by Toscana virus.

Material and methods

Sand fly collection and specimen treatment

Sand flies were collected by using CDC light traps overnight during sand fly season between June and October 2018, in seven localities in endemic (Chichaoua and Al Haouz Provinces) and non endemic area (Marrakech prefecture) of human cutaneous leishmaniasis, central Morocco [22].

Using 8 traps; for every locality, male and female sand flies were collected. Males were dissected and identified individually at species level according to morphological characters; particularly aedeagus shape [23]. Female sand flies were pooled based on trapping origin with up to 50 specimens per pool, and were stored in – 80 until biomolecular testing. Since the species identification was an invasive procedure which would have required dissection, to avoid virus or RNA degradation, females were directly tested for the TOSV presence without species identification [24].

TOSV testing

TOSV detection was carried out by real time PCR test. RNA was extracted from ground female sand fly pools using the One for All – Qiagen (Germany) RNA/DNA extraction kit on a semi-automated extractor system (Bio-Sprint 96 – Qiagen, Germany) according to the manufacturer's instructions, retro transcribed and subjected to real-time polymerase chain reaction (RT-PCR). TOSV detection in sand fly pools was carried out on CFX 96 Thermal cycler (BIO-RAD) according to Pérez-Ruiz et al. [25] protocol.

Data analysis

For investigated places, species alpha diversity with three indices were calculated using the data of the collected males in Microsoft office Excel 2010. Shannon–Weiner index (H) was calculated according to this equation: $H = - \sum_{i=1}^s p_i \ln(p_i)$.

Where p is the proportion (n/N) of individuals of one particular species found (n) divided by the total number of individuals found (N), $\ln$ is the natural logarithm, Σ is the sum of the calculations, and s is the number of species [26].

Evenness index (E) was calculated by dividing H index by natural logarithm ($\ln$) of the number of collected species (s) [26].

Species richness was estimated by calculating of Margalef's index (DMg) [26] through this equation: $(DMg) = (s - 1) / \ln N$.

To calculate the Synanthropic Index (SI), the following formula was used: $(SI) = (2a + b - 2c) / 2$ where:

- a represents the percentage of individuals of a given species collected in the urban area;
- b stands for the percentage of individuals of this same species collected in the rural area; and
- c represents the percentage of individuals of this species collected in the forest area.

The Synanthropic Index of Nuorteva (SI) ranges from +100 to –100. Negative values indicate aversion to the human environment while positive values represent a greater preference for anthropogenic environments [27]. This index was calculated for species that presented $n > 10$ and present in all biotopes needed for the calculation of the index.

Correlations were tested with the unilateral Spearman's rank correlation test in R 3.5.1 [28].

Results

A total of 6798 specimens (3558 females and 3240 males) were collected during the period from June 26th to October 11th, 2018 in seven localities from Chichaoua, Al Haouz and Marrakech.

The morphological identification performed on male sand flies allowed us to identify 8 species, divided into 2 genera: *Phlebotomus* and *Sergentomyia*. Five species (*P. sergenti*, *P. longicuspis*, *P. papatasi*, *S. minuta* and *S. fallax*) were common in all environments, with varied abundance (Table 1).

P. sergenti was the most common species (43,12%), followed by *P. papatasi* (18,89%), *P. longicuspis* (13,43%), *P. perniciosus* (8,46%), *S. minuta* (7,72%), *S. fallax* (6,45%), *S. dreyfussi* (1,17%) and *P. alexandri* (0,77%). *P. sergenti* was the dominant species in four localities that had an altitude above 912 m: Lalla Laaziza, Aderdor, Imintanout and Ourika Valley. *P. longicuspis* was recorded as the most abundant sand fly in the Ait Ourir locality. *P. papatasi* was the most abundant species in two localities: Marrakesh and Ait Faska. The details of sampling in each site are shown in Table 1. The sex ratio favored females in most stations with the exception of Aderdor locality (M/F = 1:1) (Table 3). However, the species richness changed according to locality, being higher in Ourika valley and lower in Marrakech (Table 2). The analysis of alpha biodiversity phlebotominae sand flies showed that Ourika Valley, a forest, had the highest diversity according to the DMg index ($H = 1.535$, $E = 0.738$, $DMg = 1.321$). The details of alpha biodiversity indices at different altitudinal ranges are given in Table 2. We observed that the richness, diversity and evenness were maximal at 912–1456 m of altitude and lower in areas between 399 and 713 m of altitudes (Table 2). The number of *P. sergenti*, *P. longicuspis* and *P. perniciosus* showed a positive

Table 1
Sand fly species composition per locality based on the dissection and identification of male.

Trapping locality	Co-ordinates	Region	<i>Phlebotomus papatasi</i>	<i>P. sergenti</i>	<i>P. alexandri</i>	<i>P. longicuspis</i>	<i>P. perniciosus</i>	<i>Sergentomyia fallax</i>	<i>S. minuta</i>	<i>S. dreyfussi</i>	Total
Aderdor	31°03'N8°44'W	Chichaoua ^a	11	633	22	204	140	22	43	0	1075
Lalla Laaziza	31°04'N8°42'W		39	236	0	54	51	57	26	32	495
Imintanout	31°10'N8°51'W		231	277	0	21	17	3	11	0	560
Ait Ourir	31°33'N7°40'W		55	40	0	73	34	33	31	0	266
Ait Faska	31°30'N7°43'W	Al Haouz ^a	62	49	0	48	27	25	37	0	248
Ourika Valley	31°22'N7°46'W		54	93	3	9	5	18	12	6	200
Marrakech	31°35'N8°0'W	Marrakech	160	69	0	26	0	51	90	0	396
Total			612	1397	25	435	274	209	250	38	3240
%			18,89	43,12	0,77	13,43	8,46	6,45	7,72	1,17	100

^a Endemic area of human cutaneous leishmaniasis.

Table 2
The results of alpha biodiversity of phlebotominae sand flies in seven localities investigated in center of Morocco.

Biotope	Area	Altitude	Total of specimen (N)	Sex-ratio (M/F)	Number of species (S)	Shannon–Weiner biodiversity index (H)	Evenness (E)	Richness (DM _g)
Human house	Rural	1496	238	1,1	7	1,865	0,958	1,276
Animal shelters			1812		7	1,399	0,719	0,862
Human house	Rural	1148	680	1,41	7	1,431	0,735	0,919
Animal shelters			164		7	1,605	0,825	1,1764
Animal shelters	Urban	912	4795	0,13	6	1,61	0,898	0,589
Human house	Urban	632	560		6	1,74	0,97	0,7901
Animal shelters	Rural	713	390	0,9	6	1,72	0,93	0,951
Human House			133		6	1,74	0,94	1,242
Forest	Rural	1367	485	0,7	8	1,535	0,738	1,321
Human house	Urban	399	691	1,3	5	1,49	0,931	0,668

Table 3
Synanthropic index for male sand flies (n > 10) in three areas (Natural, rural, urban).

Species	Synanthropic index (SI)	The significance of the different degrees of SI	Values
<i>P. papatasi</i>	+ 12,6	preference for anthropogenic environments	+ 20 0
<i>P. sergenti</i>	+ 10,3	preference for anthropogenic environments	+ 20 0
<i>P. longicuspis</i>	+ 13,4	preference for anthropogenic environments	+ 20 0
<i>S. minuta</i>	+ 8,1	preference for anthropogenic environments	+ 20 0
<i>S. fallax</i>	+ 1,1	preference for anthropogenic environments	+ 20 0

Table 4
Number of specimen's females tested for TOSV by trapping site.

Trapping locality	Tested specimen (n)	Total pools tested	Total positive pools
Aderdor	975	20	0
Lalla Laaziza	270	6	1
Imintanout	1633	33	0
Ait Ourir	150	3	0
Ait Faska	135	3	0
Ourika valley	100	2	0
Marrakech	295	6	0
Total	3558	73	1

correlation (($r_s = 0.997$, $P = 0.0003$), ($r_s = 0.999$, $P = 0.0003$) and ($r_s = 0.986$, $P = 0.01$), respectively) with altitude, which is not the case for the other species. Furthermore, we observed that some species were more abundant in human dwellings than in animal shelters, as the cases of *P. papatasi*, *P. sergenti* and *S. minuta* (Tables 2 and S1).

The synanthropic index of *P. sergenti*, *P. papatasi*, *P. longicuspis*, *S. minuta* and *S. fallax* were calculated. Estimated synanthropic indices for these species were between + 1.1 and + 12.6 (Table 3). *S. fallax* had the lower synanthropic index in comparison with other species who appear more attracted to humans.

Among the 73 female pools examined, one pool resulted positive for TOSV (Table 4). The TOSV positivity was detected in one of the

seven investigated study sites. TOSV RNA was detected in one pool (1/6) in the Lalla Laaziza locality in a human dwelling biotope from the Chichaoua province (Table 3), as it is shown in the amplification plot of the TOSV RT-PCR in additional material (S2). The sand fly fauna investigations in the Lalla Laaziza locality linked with positivity in this study show that *P. sergenti* were the most abundant species in this area (47,6%), followed by *S. fallax* (11,5%), *P. longicuspis* (10,3%), *P. perniciosus* (9,7%), *P. papatasi* (7,9%), *S. dreyfussi* (6,5%) and *S. minuta* (5,3%). This finding shows that TOSV is circulating in Chichaoua province; an area in which this vector-borne pathogen was previously unreported (Fig. 1).

Discussion

To detect the circulation of TOSV in central Morocco, sand flies were collected from seven localities of three provinces in central Morocco; and females were submitted to TOSV RT-PCR. Two of the three provinces are active foci of CL sustained by *L. tropica*: Chichaoua [29] and Al Haouz [30], with sporadic cases of visceral leishmaniasis (VL) caused by *L. infantum* [22].

Species composition shows that *P. sergenti*, *P. papatasi*, *P. longicuspis*, *S. minuta* and *S. fallax* were the most frequently collected species in the sampling sites (Table 2). Some studies reported the high abundance of those species in Morocco [31–33]. Several investigations have shown that altitude affects the spatial distribution and density of the sand fly fauna [32,34]. We noted a positive

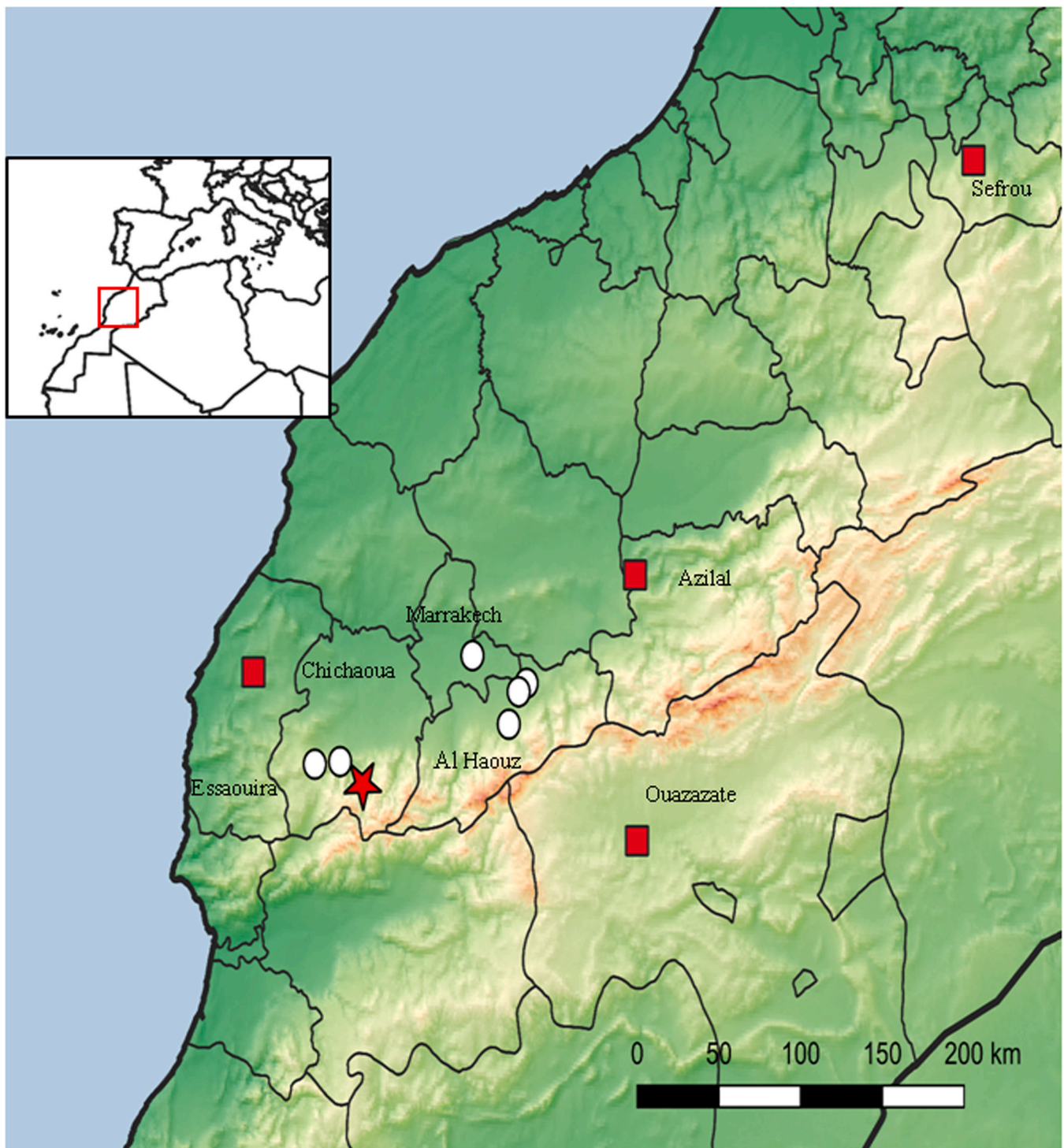


Fig. 1. Map with locations of surveyed sites (white circle), position of area with TOSV circulation in Morocco (red square) and our TOSV detection (red star).

correlation with altitude for *P. sergenti*, *P. longicuspis* and *P. perniciosus* (Table 2). This finding confirms the results of previous studies in the High-Atlas Moroccan Mountains [32,33]. These species are known to be dominant in anthropic environments and are easily collected in intradomestic biotopes [30,35], as confirmed in our study by the obtained synanthropic indexes. Between these species, *P. sergenti* was reported as common vector of anthroponotic cutaneous leishmaniasis (ACL) due to *L. tropica* and TOSV in Morocco [15,16]; *P. longicuspis* as vector of VL due to *L. infantum* [36] and suspected vector of TOSV in Morocco [37]; *P. papatasi* as the common

vector of zoonotic cutaneous leishmaniasis (ZCL) due to *L. major* in Morocco [38]. Other sand fly species such as *S. dreyfussi* and *S. antennata*, *P. alexandri* were found, but less frequently. In Morocco, the distribution of sand flies was primarily influenced by the bioclimatic factors [39–41]. Boussaa and Boumezzough [34] identified the effect of soil texture on sand fly habitats. The differences in biodiversity, richness, and evenness components of Phlebotominae sand flies are attributed greatly to abiotic factors, such as altitude. Our results showed that the biodiversity index and richness were higher in highland areas (912 m above sea level) than in the plain area. These

results are in good agreement with previous investigations in Morocco [42]. We also observe that abundance of sand flies are higher in animal shelters than in domestic biotopes. In contrast, species richness and biodiversity are important in domestic biotopes. Several factors can justify this finding such as altitude, habitat and host preferences [43,44].

Our molecular investigations showed the presence of TOSV in Lalla laaziza locality in Chichaoua province, where the virus was detected in this study. This province is an emerging epidemic focus of ACL due to *L. tropica* [45]. In Chichaoua province, VL is sporadic, while human CL poses a significant public health problem, and in fact 825 cases were officially reported in 2016 [45]. This result suggests the active circulation of TOSV in Morocco, particularly in central provinces of the country. This is also demonstrated by other investigations which recorded the virus across Azilal, Sefrou, Essaouira, and Ouarzazate, areas known as endemic areas of CL [46]. To preserve their integrity, tested females were not identified at species level. However, male identification gave a precise picture of the sand fly fauna in the Lalla Laaziza locality and highlighted *P. sergenti* as the most abundant species in this site (47%), followed by *P. perniciosus* (10.3%) and *P. longicuspis* (9.7%). Interestingly, all these three species were tested TOSV positive in Morocco [46]. Regarding TOSV detections in Morocco, the virus detected and isolated in an area with a species composition known *P. sergenti* as the most abundant species in Azilal and Essaouira provinces, *P. longicuspis* in Sefrou province and *P. papatasi* in Ouarzazate province [46]. Thus, TOSV is typically transmitted by *P. perniciosus* and *P. perfliewi* sand flies in southern European countries [12]. These results suggest the ability of the virus to infect a wide range of sand flies, rather than a single species as the main vector.

With regard to altitude, TOSV positivity was previously reported in the region of Draa El Mizan of Algeria at an average altitude of 380 m [47]. We detected TOSV in this study at an altitude of 1148 m at Lalla laaziza, in the same range reported in other studies conducted in Louata and Azilal; 950 m and 1351 m respectively [46] (Table 1).

In Morocco, there is a scarcity of data on diseases linked to phlebovirus infections. The circulation of TOSV in sand flies encourages consideration of this virus as an etiological agent of meningitis, encephalitis and meningoencephalitis in humans. In addition, considering that the results of previous studies showed the presence of phlebovirus antibodies in humans, this indicates a likely circulation of phleboviruses in the Moroccan population of sand flies [48,49]. The TOSV-positive pool was collected in a human dwelling, and people in Morocco had recorded antibodies against the Naples and Sicilian virus [14,48]. Consequently, potential involvement of vertebrates in the phlebovirus life cycle could be hypothesized. This has to lead future research to investigate the distribution of the TOSV in Morocco, in sand flies and in the human population, through direct tests or evaluating the immune response to infection; we should also consider the co-infection of sand flies with both *Leishmania* and phleboviruses, and as possible relevant human pathological interaction. The TOSV-positive pool was collected in a humandwelling, and people in Morocco had recorded antibodies against the Naples and Sicilianvirus [14,48,50].

Conclusion

We found that TOSV is circulating in Chichaoua province, an area endemic for leishmaniasis in Morocco. This finding calls for further investigation of potential effects of TOSV on human health in this area, since no epidemiological data of human infections is available. This finding highlights the importance of surveying the presence of TOSV in sand flies to characterize the geographical circulation of the virus.

CRedit authorship contribution statement

A.B., S.B., M.C., M.D., M.H., A.L. and M.D. conceived of and designed the study; M.D., M.C., P.B., D.L., D.T., G.R. and S.B. performed the experiments and analyzed the data; M.D., S. B., and M.C. wrote the paper; M.D., M.C., S.B., P.B., M.D., M.H., A.B., A.L., D.T. and D.L. were involved in the interpretation of the results and critical reading of the manuscript. All authors have read and agreed to the published version of the manuscript.

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Ethics approval

Not applicable because there are no animals and humans used in this study.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Code availability

Not applicable.

Availability of data and material

Not applicable.

Conflicts of interest/Competing interests

There are no conflicts to declare.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2022.03.007.

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